Singapore School Kinderland Secondary-2

Computer Science — Term 2 Comprehensive Study Material & Syllabus Guide

Cambridge IGCSE & O Level Computer Science (Syllabus 0478 / 2210)

Prescribed Textbook: Cambridge IGCSE Computer Science (2nd Edition)

Chapters Covered: Chapter 1.3 (Pg 32–44) & Chapter 2.2, 2.3 (Pg 54–74)

Academic Level: Secondary 2 (Grade 8)

Examination Scope: Class Test 1, Class Test 2 & Term 2 Final Exams

MODULE 1: CHAPTER 1.3 DATA STORAGE & FILE SIZE CALCULATIONS (TB PG 32–40)

1.3.1 Measurement of Data Storage & Prefixes (TB Pg 32–34)

Fundamental Storage Units:

- **Bit (b):** Smallest unit of data representation, binary 0 or 1.
- **Nibble:** A group of 4 bits (e.g. 1010).
- **Byte (B):** A group of 8 bits. Capable of representing one ASCII character.

Metric (SI) Prefixes (Powers of 10): Used by storage manufacturers: 1 Kilobyte (KB) = 1,000 Bytes (10^3).
IEC Binary Prefixes (Powers of 2): Standard computer internal architecture: 1 Kibibyte (KiB) = 1,024 Bytes (2^{10}).

Unit Name	IEC Binary Symbol	Exact Power of 2 Value	Traditional SI Equivalent
Kibibyte	KiB	1,024 Bytes (2^{10})	1 KB = 1,000 Bytes (10^3)
Mebibyte	MiB	1,024 KiB ($2^{20} = 1,048,576$ B)	1 MB = 1,000 KB (10^6)
Gibibyte	GiB	1,024 MiB ($2^{30} = 1,073,741,824$ B)	1 GB = 1,000 MB (10^9)
Tebibyte	TiB	1,024 GiB (2^{40})	1 TB = 1,000 GB (10^{12})

1.3.2 Image File Size Calculations (TB Pg 35–37)

Bitmapped images are made up of individual picture elements called **pixels**. The size of an uncompressed image depends on two factors:

- **Image Resolution:** Number of pixels in width × height (e.g., 1024 × 768 pixels).

- **Colour Depth:** Number of bits used to represent the color of each pixel (e.g., 1 bit = 2 colors, 8 bits = 256 colors, 24 bits = 16.7 million true colors).

> **Formula: Image File Size (in bits) = Image Width (px) × Image Height (px) × Colour Depth (bits)**
> **In Bytes: File Size (Bytes) = File Size (bits) ÷ 8**
> **In Kibibytes (KiB): File Size (KiB) = File Size (Bytes) ÷ 1024**

Worked Example: Calculate the file size of a photograph measuring 800 × 600 pixels with 24-bit True Colour depth.
1. Total bits = 800 × 600 × 24 = 11,520,000 bits.
2. Total Bytes = 11,520,000 ÷ 8 = 1,440,000 Bytes.
3. Total KiB = 1,440,000 ÷ 1,024 ≈ **1,406.25 KiB** (≈ 1.37 MiB).

MODULE 1 (CONTINUED): SOUND MATH & COMPRESSION ALGORITHMS (TB PG 38–44)

1.3.3 Sound File Size Calculations (TB Pg 38–40)

Sound is analogue by nature. To store it digitally, sound waves are sampled at regular intervals:

- **Sample Rate:** Number of sound samples taken per second, measured in Hertz (Hz) or kHz (e.g. CD quality = 44,100 Hz).
- **Sample Resolution:** Number of bits allocated per sample (bit depth, e.g. 16 bits).
- **Duration:** Total time of audio in seconds.
- **Channels:** Mono = 1 channel; Stereo = 2 channels.

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Formula: Sound File Size (bits) = Sample Rate (Hz) × Sample Resolution (bits) × Duration (s) × Channels
In Bytes: File Size (Bytes) = File Size (bits) ÷ 8
In Mebibytes (MiB): File Size (MiB) = File Size (Bytes) ÷ (1024 × 1024)
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Worked Example: A 30-second stereo audio track sampled at 44.1 kHz (44,100 Hz) with 16-bit resolution:
1. Bits = 44,100 × 16 × 30 × 2 = 42,336,000 bits.
2. Bytes = 42,336,000 ÷ 8 = 5,292,000 Bytes.
3. MiB = 5,292,000 ÷ 1,048,576 ≈ **5.05 MiB**.

1.3.4 Lossy vs Lossless Compression (TB Pg 41–44)

Feature	Lossy Compression	Lossless Compression
Data Removal	Permanently removes redundant or imperceptible data (e.g. sounds human ears cannot hear).	No data is lost. Original file can be decompressed back to 100% exact original form.
File Size Reduction	Massive reduction (up to 90% smaller).	Moderate reduction (usually 30% to 50% smaller).
Common Formats	JPEG (images), MP3/AAC (audio), MP4 (video).	PNG, SVG (images), FLAC, ALAC (audio), ZIP, TXT.
Suitable Uses	Web streaming, photos where slight quality drop is acceptable.	Text files, executable programs, spreadsheets, medical scans.

1.3.5 Run-Length Encoding (RLE) Algorithm (TB Pg 43–44)

Concept: RLE reduces the size of repeated sequences of identical data (runs). It stores the **count (run length)** followed by the **value**.
• Uncompressed Text: `AAAAABBBCCCCCDD` (15 bytes)

- RLE Encoded: `05 A 03 B 05 C 02 D` (8 bytes / pairs) → **46.7% Compression Ratio!**
- *Limitation:* If data has no consecutive repeating runs (e.g. `ABCDEF`), RLE will actually increase the file size!

MODULE 2: CHAPTER 2.2 METHODS OF ERROR DETECTION (TB PG 54–62)

During data transmission across telecommunication cables or wireless links, electrical interference, attenuation, and noise can cause bits to flip ($0 leftrightarrow 1$). Five standard methods detect these errors:

2.2.1 Parity Checks & Parity Blocks (TB Pg 54–57)

- **Parity Bit:** An extra bit added to the 7-bit byte to ensure the total number of 1s is either **Even** (Even Parity) or **Odd** (Odd Parity).
- *Single Parity Limitation:* If two bits change during transmission, the parity count remains identical and the error goes completely undetected!
- **Parity Block Matrix:** Combines horizontal byte parity bits with a vertical **Parity Byte** at the bottom. The intersection of the corrupted row and corrupted column pinpoints the exact bit that flipped, enabling automatic correction without retransmission!

2.2.2 Checksum Algorithm (TB Pg 58–60)

> 1. Sender passes data blocks through an algorithm to compute a checksum value.
> 2. Checksum is appended to the packet and sent across the network.
> 3. Receiver recalculates the checksum using the same algorithm on arrival.
> 4. If the computed checksum matches the received checksum, data is error-free; otherwise, a re-transmission request is sent.

2.2.3 Echo Check (TB Pg 59–60)

The receiving computer transmits the entire received data block back to the sending computer ("echo"). The sender compares the echo with original. *Major Drawbacks:* Consumes double bandwidth, and cannot determine whether corruption occurred on the way forward or on the way back.

2.2.4 Check Digits — Modulo 11 & ISBN-13 (TB Pg 60–61)

Calculated at the end of identification numbers (barcodes, account numbers) to catch human manual data entry errors:

- **Transcription errors:** Mistyping a single digit (e.g. typing 3 instead of 8).
- **Transposition errors:** Swapping two adjacent numbers (e.g. typing 78 instead of 87).

2.2.5 Automatic Repeat Requests (ARQ) (TB Pg 61–62)

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Protocol Flow:
1. Sender transmits packet and starts a Timeout Timer.
2. If receiver receives packet intact, it sends back a Positive
Acknowledgement (ACK).
3. If corrupted, receiver sends Negative Acknowledgement (NAK) → sender
retransmits immediately.
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> **4. If no response arrives before the timer expires (Timeout), the packet is automatically re-sent until successful or limit reached.**

MODULE 3: CHAPTER 2.3 SYMMETRIC & ASYMMETRIC ENCRYPTION (TB PG 63–74)

Encryption transforms readable **Plain Text** into unintelligible **Cipher Text** using an algorithm and cryptographic key to protect confidentiality.

2.3.1 Symmetric Encryption (TB Pg 63–67)

- Uses the **same shared secret key** for both encryption and decryption.
- *Advantage:* Very fast and computationally efficient; ideal for encrypting large databases and hard drives.
- *Critical Vulnerability (Key Distribution Problem):* The sender must transmit the secret key to the receiver. If intercepted by an eavesdropper, all future communications are compromised.

2.3.2 Asymmetric Encryption (TB Pg 68–74)

Uses mathematically linked pairs of keys: a **Public Key** (openly published to the world) and a **Private Key** (kept strictly secret by the owner).

Operation	Key Used for Encryption	Key Used for Decryption / Purpose
Confidential Messaging	Recipient's **Public Key**	Recipient's **Private Key** (Only the recipient can open and read the message).
Digital Signature (Authenticity)	Sender's **Private Key**	Sender's **Public Key** (Proves message genuinely originated from sender and has not been tampered with).

2.3.3 Hashing & SSL/TLS Digital Certificates (TB Pg 70–74)

- **Hashing Function:** A one-way mathematical algorithm (e.g. SHA-256) that converts input data into a fixed-length hash digest. Impossible to reverse back into plain text. Used for storing passwords securely and verifying file integrity.
- **SSL / TLS Digital Certificates:** Issued by trusted Certificate Authorities (CAs). Verifies the authentic identity of a website, displays the padlock in browser URL, and exchanges asymmetric keys to establish secure encrypted symmetric sessions (HTTPS).

> 💡 **Exam Preparation Tip for Secondary 2 Students:**
> • Always state formulas clearly in file size calculations before writing numerical steps.
> • Remember that $1 ext{ KiB} = 1024 ext{ Bytes}$, whereas $1 ext{ KB} = 1000 ext{ Bytes}$. Cambridge penalizes confusing these prefixes!
> • For Parity Blocks, check both rows and columns: the intersection points directly to the erroneous bit.